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Flash Sync Speed **[10 pts]**

Q1. Illuminated Area

Step 1: the swept height. Flip the sensor upside down so the illuminated band sits at the top. By the illumination lemma,

$$s(t^*) = 24\,\text{mm} \times \frac{\text{shutter speed}}{\text{flash sync time}} = 24\,\text{mm} \times \frac{1/200\ \text{s}}{14.73\,\text{ms}} = 24 \times \frac{5}{14.73} \approx 8.147\,\text{mm}.$$

Step 2: the paper's vertical extent. Both pieces are elevated 2 mm off the board, and the disk of radius 10 mm centred at $y = 12$ also has its lowest point at $y = 2$. Hence the paper occupies $y \in [2, 22]$, and its overlap with the illuminated band $[0, 8.147]$ is

$$y \in [\,2,\ 8.147\,].$$

Step 3: integrate. At each height y, the horizontal width of a strip is proportional to the square of its distance from the thin tip, so the three contributions are:

Strip 1 (broad head down). Width $\dfrac{(22-y)^2}{50}$:

$$A_1 = \int_2^{8.147} \frac{(22-y)^2}{50}\,\mathrm{d}y = \frac{(20)^3 - (22-8.147)^3}{150} \approx 35.61\ \text{mm}^2.$$

Strip 2 (thin head down). Width $\dfrac{(y-2)^2}{50}$:

$$A_2 = \int_2^{8.147} \frac{(y-2)^2}{50}\,\mathrm{d}y = \frac{(8.147-2)^3}{150} \approx 1.55\ \text{mm}^2.$$

Disk. Width $2\sqrt{100-(y-12)^2}$:

$$A_3 = \int_2^{8.147} 2\sqrt{100-(y-12)^2}\,\mathrm{d}y \approx 81.96\ \text{mm}^2.$$

(The antiderivative is $(y-12)\sqrt{100-(y-12)^2} + 100\arcsin\frac{y-12}{10}$; the numeric value follows from a CAS.)

Total illuminated area:

$$\boxed{A = A_1 + A_2 + A_3 \approx 35.61 + 1.55 + 81.96 = 119.1\ \text{mm}^2}.$$